AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X

MODEL EXAMINATION PAPER 2027 AND ONWARDS

Computer Science Paper II

Time: 1 hour 50 minutes Marks: 25

This model paper aims to represent the table of specification with regards to the topics/ content and cognitive demand only.

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INSTRUCTIONS:

1. Check your name and school information above. Sign if the information is correct.

I agree that this is my name and school. **Candidate's Signature**

2. Read each question carefully.
3. There are SEVEN questions. Answer ALL questions. Questions 6 and 7 each offer TWO choices. Attempt any ONE choice from each.
4. When answering the questions:
   - Use a black pen/ pointer to write your answers. DO NOT write your answers in pencil.
   - Use a black pencil for diagrams and drawings. DO NOT use coloured pencils, markers or highlighters.
   - DO NOT use staples, paper clips, glue, correcting fluid, or ink erasers.
     If you make a mistake, cross it out neatly and write the correct answer beside it.
   - Write your answers only in the allocated space provided for each question.
   - DO NOT write outside the answer space. Answers written outside the allocated space will not be marked.
5. The marks for each question are shown in brackets ( ) next to the question number. Use this information to manage your time efficiently.
6. DO NOT mention your name or school name anywhere inside the paper.
7. You may use a simple calculator if you wish.

Q.1. (Total 3 Marks)

Solve the following using the binary number system.

i. $1010110_2 + 110101_2 = ?$
ii. $1110011_2 – 1011011_2 = ?$
iii. $101101_2 * 101_2 = ?$

Q.2. (Total 3 Marks)

Write an algorithm using a for loop to print the first 10 natural numbers.

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Q.3. (Total 2 Marks)

Describe program syntax and program semantics.

Q.4. (Total 2 Marks)

Write JavaScript code to declare and initialise a one-dimensional array that contains mixed data types (e.g., number, string, boolean, and null).

Q.5. (Total 3 Marks)

Write a JavaScript code that uses the prompt( ) function to read a number from the user and display whether it is even or odd.

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Q.6. (Total 6 Marks)

EITHER

a. Write JavaScript code to perform the following task:

- ask the user to enter a menu option number using an input statement.
- use a switch statement to display:

  "Add Record" for option 1
  "Update Record" for option 2
  "Delete Record" for option 3
  "View Record" for option 4
- display "Invalid Option Selected" if the entered value does not match any case.
- display the output using JavaScript.

OR

b. Write JavaScript code to perform the following task using a for loop:

- ask the user to enter a positive integer n representing the number of students.
- ask the user to enter marks for each student.
- calculate the total marks and average marks.
- display the result in the format:

  "Total Marks: total"
  "Average Marks: average"
- if the user enters a number less than 1 for n, display "Invalid number of students".

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Q.7. (Total 6 Marks)

EITHER

a. Ayesha, a high school student, becomes the target of cyberbullying after someone shares edited photos of her in a group chat. Soon, others began posting hurtful comments and mocking her on social media.

Based on the given situation, analyse any SIX different ways through which cyberbullying can affect Ayesha's emotional, social, and academic well-being.

OR

b. Analyse any SIX different impacts of Artificial Intelligence (AI) in the fields of robotics, military and aviation, mechanical engineering, and AI-assisted surgery.

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END OF PAPER

Please use this page for rough work

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